Public Transp (2009) 1: 103–120
DOI 10.1007/s12469-008-0007-0

ORIGINAL PAPER

Driver scheduling problem modelling

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Published online: 15 November 2008

Abstract The Drivers Scheduling Problem (DSP) consists of selecting a set of duties for vehicle drivers, such as bus, train and boat drivers or plane pilots, for the transportation of passengers or goods. This is a complex problem as it involves several constraints related to labour and company rules and may also entail different evaluation criteria and objectives. The ability to develop an adequate model for this problem, which can represent the real problem as closely as possible is an important research area.

In this paper we present new mathematical models for the DSP which embody the very complexity of the drivers scheduling problem, besides demonstrating that the solutions generated by these models can easily be implemented in real situations.

On the strength of extensive passenger transportation experience in bus companies in Portugal, we propose and test new alternative models to formulate the DSP. These models are based on Set Partitioning/Covering models. Moreover, they also take into account the bus operator issues and the user's standpoint and environment.

Keywords Drivers · Duties · Modelling

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1 Introduction

The Drivers Scheduling Problem (DSP) consists of selecting a set of duties for vehicle drivers, such as bus, train and boat drivers or plane pilots, for the transportation of passengers or goods. This is a complex problem as it involves several constraints related to labour and company rules and can also present different evaluation criteria and objectives. The ability to develop an adequate model for this problem, which can represent the real problem as closely as possible is an important research area.

The main objective of this research work is to propose new mathematical models for the DSP that represent the overall complexity of the drivers scheduling problem, besides demonstrating that the solutions generated by these models can easily be implemented in real situations. This issue has been recognized by several authors and constitutes an important problem in Public Transportation.

The most well-known, general formulation for the DSP is a Set Partition/Set Covering Model. However, to a large extent these two models simplify some of the specific business aspects and issues of real problems. This makes it difficult to use such models as automatic planning systems because the schedules obtained must be modified manually in order to be implemented in real situations. As mentioned by Borndörfer et al. (2006) this is a significant problem in the area of Public Transportation. The authors call for studies on this issue and say "The challenge here is not yet mathematical; it is to find adequate O.R. models." The main purpose of our work is to contribute to the discovery of better DSP models.

On the strength of extensive passenger transportation experience in bus companies in Portugal, we propose new alternative models to formulate the DSP. These models are also based on Set Partitioning/Covering Models. However, they take into account the bus operator issues, as well as the user's standpoint and environment.

We pursue the classical steps employed by Operations Research Methodology. All the processes are performed with the close participation and involvement of the final users from different transportation companies. The planners' opinion and major criticisms serve to improve the model proposed in a continuous enrichment process.

The ultimate objective is to come up with a model that can be incorporated into an information system for use as an automatic tool to produce driver schedules. Therefore, the criteria for evaluating the models consists in the capacity to generate real and useful schedules that can be implemented with few manual adjustments or modifications. We have considered the following features as measures/measurements of the quality of the model: simplicity, solution quality and applicability.

We tested the alternative models using a set of real data acquired from several different transportation companies. We then analyzed the optimal schedules obtained with respect to the applicability of the solution to the real situation. To do this, the schedules were analyzed by the planners to determine their quality and applicability. The main result of this work lies in the proposition of new mathematical models for the DSP that provide an enhanced representation of the realities of the passenger transportation operators and lead to better schedules that can be implemented directly in real situations.

2 Drivers scheduling problem

To evaluate different alternative models for the DSP we have followed the seven steps of the Operations Research Methodology (Winston 1994). They are as follows: Identify the Problem; Understand the System; Formulate a Mathematical Model; Verify the Model; Select the Best Alternative; Present the Results of the Analysis and Implement and Evaluate. All the processes involve the close participation and involvement of the final users from different transportation companies, the main objective being to obtain realistic solutions from the models.

In the first step, *Identifying the Problem*, we start by defining the main objectives of the research, which are: To develop mathematical models for the DSP that most closely represent the realities of passenger transportation operators. These models will be incorporated in a Decision Information System, and we hope the models will produce schedules that can be implemented directly in real situations. In short, our objective is to take a closer look at the DSP and obtain models and solutions that make business sense to the users.

Although in theory an analyst would hope to include the broad issues and constraints of the problem in the proposed model, a model cannot include every aspect of a situation. A model is always an abstraction that is necessarily a simplification of the real situation; elements that are irrelevant or unimportant to the problem are ignored. The elements considered in the model should leave sufficient detail so that the solution obtained has value with regard to the original problem which, in this case is the real planning process. Models must be both tractable, capable of being solved and valid, representative of the original situation. These dual goals are often contradictory and not always attainable. It is generally true that the most powerful solution methods can be applied to the simplest or most abstract model. Therefore, we are looking for simple but applicable models with high quality solutions.

In the next phase *Understand the System*, we define the main concepts and collect adequate data. The main concepts are explained below.

The base of the DSP is the Vehicle Schedule which aggregates the need for a driver or crew member for each of several vehicles. A *Block* is the work of each vehicle from the time it leaves the depot, or the place were it is parked, until it returns to the same depot or location, in a particular period.

A *Relief Point* is a place where it is possible to change the driver who is at the wheel of a vehicle. A *Relief Opportunity* is a pair composed of the place and hour where and when it is possible to change the driver who is driving a vehicle. A *Piece of work* is the period between two consecutive relief opportunities.

The DSP definition starts with the division of all blocks of vehicles of a schedule into pieces of work. If the difference between two non-consecutive pieces of work is smaller than the break duration it is considered to be a *Little Interruption*. This period of time is usually paid time as the driver is in the vehicle, but he or she is not driving. A *Stretch* is a consecutive period of vehicle driving, formed by several consecutive pieces of work or a set of pieces of work separated by short interruptions.

A *Duty* for a driver is a set of stretches that involve the bus driving periods, the break and meal periods and the travel of the driver, walking or driving, to the point where the duty or the stretch starts. The duties are defined by a large set of parameters that reflect labour rules, security procedures and planning strategies. These parameters are different and depend on the type of duty, and on the number of stretches and breaks permissible within a duty. A *Feasible Duty* is a duty that meets all the parameters and restrictions that define its type.

The DSP can be defined as follows: Given a set of pieces of work, that result from the division of blocks of a vehicle schedule, obtain the set of feasible duties that guarantee the driving of all vehicles with a minimum cost. In this phase, besides describing the DSP we also looked for issues and rules associated with the problem. These will be described in the course of the next session.

3 SPP/SCP models

In the three sections that follow we *Formulate a Mathematical Model*. We start by presenting the most traditional models for the DSP and discuss the advantages and disadvantages of these models. We then argue how the users evaluate the solutions of the models in Sect. 4, and in the subsequent section we present alternative models for the DSP. These are based on the criticisms and comments obtained by the planners. In the interests of clarity and given the importance of this phase, we have decided to present it in three sections.

One of the most common formulations of the DSP is the Set Covering or Set Partitioning Model (SPP/SCP). In the SPP model each piece of work is covered by only one duty. In the SCP model it is possible to have more than one duty covering each piece of work.

In the SPP/SCP models there is a set of pieces of work or rows that needs to be covered and a set of previously pre-defined feasible duties or columns that covers specific pieces of work. The DSP resolution, based on these two models, involves selecting the feasible duties which guarantee that there is one (SPP) or more (SCP) duties covering each piece of work, thus minimizing the total cost of the final schedule.

Consider the SPP/SCP models, such that a row represents a piece of work and $M = \{1, \dots, m\}$ is the set of all rows of the problem. A column represents a previously generated feasible duty and $N = \{1, \dots, n\}$ is the set of columns of the problem.

The coefficients a_{ij} ($i \in M$ and $j \in N$) of the constraints matrix are:

$$a_{ij} = \begin{cases} 1 & \text{if work piece } i \text{ is covered by duty } j; \\ 0 & \text{otherwise.} \end{cases}$$

The decision variable x_j ($j \in N$) means:

$$x_j = \begin{cases} 1 & \text{if the duty } j \text{ is in the schedule;} \\ 0 & \text{otherwise.} \end{cases}$$

A solution to the DSP formulated with SPP/SCP models is a vector of decision variables $x = (x_1, x_2, \dots, x_n)$, with the value 0 or 1 which specifies if a previously generated feasible duty is on the schedule or not.

The elements c_j $(j \in N)$, coefficients of the objective function, represent the cost of a feasible duty j.

In considering the SPP model, the DSP has the following formulation:

$$\text{(SPP)} \quad \text{MIN} \quad \sum_{j=1}^{n} c_j x_j \tag{1}$$

$$\text{s.t.:} \quad \sum_{j=1}^{n} a_{ij} x_j = 1 \quad (i \in M) \tag{2}$$

$$x_j \in \{0, 1\} \quad (j \in N) \tag{3}$$

Constraint set (2) guarantees that all pieces of work must be covered by only one duty on the schedule. The objective function corresponds to total cost minimization.

If $c_j = 1 \; \forall j \in N$ the objective function corresponds to the minimized total number of duties.

In considering the SCP model, the DSP has the following formulation:

$$\text{(SCP)} \quad \text{MIN} \quad \sum_{j=1}^{n} c_j x_j \tag{4}$$

$$\text{s.t.:} \quad \sum_{j=1}^{n} a_{ij} x_j \geq 1 \quad (i \in M) \tag{5}$$

$$x_j \in \{0, 1\} \quad (j \in N) \tag{6}$$

The constraints in set (5) are now inequalities, which allow pieces of work to be covered by more than one duty on the schedule.

With the SCP formulation the final schedule could contain some pieces of work covered by more than one duty, corresponding to an *Overcover Piece of work*. In practice this situation, usually called a *Reserve Situation*, corresponds to drivers working in place of transportation operators who temporarily have no vehicle to drive.

One of the biggest advantages of using SPP/SCP models to formulate the DSP is ones capability to define two completely different phases during the problem resolution operation; a *Generation Phase* and a *Resolution Phase*. The *Generation Phase* is the prior definition of the set of feasible duties based on the parameters defined by the planners. The *Resolution Phase* is the selection of a subset of feasible duties for the schedule, based on cost minimization. This two phased approach allows one to consider a large set of different rules when defining feasible duties. It also facilitates implementation of the solution methodology with different transportation operators or realities. When changing the bus operator, it is possible to preserve all resolution phase methodology and only adapt the generation phase. A large set of different labour rules can be used during feasible duty generation when considering SPP/SCP models. Therefore, the generation phase will obtain duties that verify all imposed legislation and labour rules. It is therefore guaranteed that any final schedule obtained in the resolution phase will allow for legal duties and requirements.

There is a huge number of different rules that a feasible duty should verify, such as:

- Minimum and maximum stretch duration;
- Minimum and maximum break duration;
- Minimum and maximum work duration;
- Minimum and maximum total duration;
- Maximum extra work duration;
- Maximum number of vehicle changes;
- Minimum driving duration of a particular vehicle.

Despite the simple nature of the general way in which the DSP is defined via the SPP/SCP models, in the real context of the daily business of Portuguese transportation operators, several particularities arise that lead to difficulties when using the simple versions of such models. These features are related to the presence of different evaluation criteria for the schedules, along with the overcover and undercover situations.

With respect to the evaluation criteria, the problems and difficulties associated with SPP/SCP models start with the need to consider the desired rules and alternative evaluation criteria when analysing final schedules. The desired rules are not imposed by legislation; it is possible to have feasible duties on the schedule that do not verify it. However implementation of such rules facilitates scheduling requirements and the daily scheduling needs and desires of workers.

Some of the possible desired rules are:

- Total (Normal Work + Extra Work) duty duration;
- Number of vehicle changes;
- Relief vehicles out of the depot;
- Duty type percentage.

The usual way to consider these rules in the SPP/SCP formulation is by reflecting them in the duty cost, which represents the real cost plus a penalty cost associated with the failure to verify these rules. Therefore, beyond representing the real cost of a feasible duty j, the cost parameter c_j also reflects the desirability of the duty with respect to the above rules. The penalties' values defined by the users should guarantee that duties with desired characteristics have less cost than the ones that do not. Despite the easy and intuitive definition of the concept of penalties, these values are very difficult to adjust. Each individual penalty is difficult to define but the adjustment of a set of different penalties at the same time, for different types of duties, is almost impossible, at least in the real situation of a transportation operator. Also, small variations of the values of the penalties lead to very different solutions obtained after solving the SPP/SCP model. The main complaint of planners is that they have difficulty in adjusting the penalties such that the cost function represents the appropriate and realistic evaluation criteria. To understand how they evaluate a solution, we show them several solutions and ask them to tell us the best ones. We came to the conclusion that they are using several evaluation criteria such as the real cost, but also resort to the number of duties, the number of trippers and the evaluation criteria related to the number and duration of over and under cover duties. In the next section we discuss these issues in detail when describing the evaluation criteria.

As for the undercover and overcover situations, the majority of planners display some flexibility in these types of situations. The ideal schedule has only one driver covering each piece of work in the problem, although in most situations this is not feasible. On the other hand, if the number of overcover pieces of work is not restricted, as in the SCP model, there can be an excess of overcover for some pieces. This leads to a schedule that cannot be implemented and the planner must change it manually. Therefore, one desired property of the "best" schedules is that in which the overcover situations are limited and some undercover situations are allowed, thus enabling the planner to enjoy a degree of flexibility to change the schedule if desired.

At the beginning of the joint work performed with these companies, we presented the results of SPP/SCP approaches to several transportation operator users. We asked these scheduling specialists to analyse the results and make suggestion for improvement. They added that they had no interest in being left with only one schedule, even if it was the optimal solution of the SPP/SCP model. In many situations the optimal solution of these model could not be directly implemented. They wanted to consider several active objectives and analyse the best solution obtained for each of them, before finally selecting one. The desired process the planners would like to follow is: firstly, to obtain a set of alternative schedules, analyse them, make some manual adjustments and then decide which final schedule should be considered for implementation.

The planners also mentioned that in general they need to perform some manual changes related to elements that are difficult to include in the model or make the model intractable. Examples of these elements include traffic changes, special pieces of work during soccer matches or other events, requests by drivers on a special day, etc. Ultimately the planners stated that by having schedules with duties not covered (undercover) and a limited number of overcovered duties their job will be facilitated when they have to modify the schedules manually. The difficulty of incorporating some specific elements during the manual changes to the optimal solution obtained from the SPP/SCP models leads to a situation in which users neither apply these models nor the associated system at all. In this situation users prefer to build the solution from scratch, in a simple, constructive way. Therefore, when using solutions via any model, one of the planners' major requirements is that the solutions obtained by the model are good enough to implement in reality, or be easy to modify so that elements not included in the model may be considered.

The SPP/SCP models display good measurements in terms of simplicity and solution quality, though they lack some aspects concerning applicability, as explained above. It is therefore the aim of this work to develop alternative models to the DSP, by incorporating the main criticisms and expectations of some Portuguese transportation operators' planners. In the next section we discuss the evaluation criteria and in the following one we shall present the alternative models.

4 Evaluation criteria

The objective function of the SPP/SCP models is to minimize cost. The way this cost function is defined has an important impact on the solutions obtained. As mentioned above, in many applications of the SPP/SCP, the cost function consists of two main parts: the real cost as a monetary value and some penalty costs reflecting the desired rules. For the planner the first cost is easy to understand, whereas the second one presents several difficulties during the cost-setting process. When working closely with the planners, the latter always comment on this issue, the reason being that on many occasions these penalties make no sense to them. They were therefore unable to assign the appropriate value to the penalties. However, the value of these penalties makes a significant impact on the optimal solutions obtained; see Portugal (2006) for details. Hence, one of the main issues on the applicability of the SPP/SCP models is the use of penalties on the objective function, and the models presented here try to overcome this difficulty. We propose a set of different evaluation criteria of the schedules related to three main objectives: total "real" cost, undercover work by/per number of pieces and undercover work duration of pieces. These evaluation criteria more closely reflect the way the planners evaluate a schedule.

Consider the following notations:

- HI_i—represents the ith piece of work starting hour ($i \in M$);
- HF_i—represents the ith piece of work ending hour ($i \in M$);
- $d_i = HF_i - HI_i$—represents the ith piece of work duration ($i \in M$);
- $w_i = \sum_{j=1}^{n} a_{ij} x_j$—represents the ith number of duties that cover piece of work i in the schedule x ($i \in M$);

The evaluation criteria of a schedule x are:

$$(1) \quad \text{Total "real" cost} = \sum_{j=1}^{n} c_j x_j.$$

Several planners consider that it is easier to adjust schedules with some undercover pieces of work than to analyse and adjust schedules with too many duties and lots of overcover pieces of work. In this situation it is feasible to have undercover pieces of work but their number should be minimal, as final schedules impose a driver to cover each piece of work.

$$(2) \quad \text{Total number of undercover pieces of work} = \sum_{i=1}^{m} \max(1 - w_i, 0);$$

$$(3) \quad \text{Total duration of undercover pieces of work} = \sum_{i=1}^{m} d_i \times \max(1 - w_i, 0).$$

The alternative models that will be defined to formulate the new proposed DSP permit undercover pieces of work; evaluation criteria (2) and (3) allow control of the number and properties of the undercover situation.

Even among transportation operators that keep drivers in reserve, the planners are alert to the difficulty of managing the situation where drivers are being paid without driving. It is therefore important to control the total number or the total duration of drivers in this situation. This is reflected in the following functions:

$$\text{(4)}\quad \text{Total number of overcover pieces of work } = \sum_{i=1}^{m} \max(\min(w_i - 1, 1), 0);$$

$$\text{(5)}\quad \text{Total duration of overcover pieces of work } = \sum_{i=1}^{m} d_i \times \max(w_i - 1, 0);$$

$$\text{(6)}\quad \text{Maximum number of duties involved in the same overcover piece of work } = \max_{i \in M}(\max(w_i - 1, 0)).$$

In the alternative models, that will subsequently be defined to formulate the DSP which admit overcover pieces of work, criteria (4), (5) and (6) allow control over the number and duration of overcover situations.

$$\text{(7)}\quad \text{Unfit } = \sum_{i=1}^{m} |w_i - 1|.$$

Because the ideal schedules have only one driver covering each piece of work of the problem, Chu and Beasley (1995) suggest that criterion (7) is a good measurement of feasibility of the constraints $\sum_{j=1}^{n} a_{ij} x_j = 1$.

$$\text{(8)}\quad \text{Number of duties } = \sum_{j=1}^{n} x_j.$$

This is one of the most important measurements of schedule quality. The personnel costs for a transportation operator are very important and one of the biggest involved in the service. The number of available drivers is not always sufficient for the daily operations. Good schedules must always contain a minimum number of duties.

(9) Trippers number.

During the planning process, some companies prefer to include the constraint that all pieces of work must be covered by some duty. However, in general this leads to unfeasible solutions or to many overcover solutions. In this situation, the general procedure is to consider *Trippers*. A *Tripper* is a duty that covers one and only one piece of work. In practice the trippers correspond to extra or voluntary work and involve additional costs. A good schedule does not have trippers. Therefore, an important objective function is to minimize the number of trippers used in a solution.

Next we combine these objective functions with some realistic constraints to develop new mathematical models for the DSP.

5 Alternative models to the DSP based on SPP/SCP

When developing a model for an operational research problem, Hillier and Lieberman (1995) proposed that it should start as a very simple version that evolves to a more complex and real one, better reflecting the managers' perspective. This process, called *Model Enrichment* was considered during this research, in addition to some modelling concept/s present in the work of Ravindran et al. (1987).

One of the most important measurements of schedule quality is the total number of duties, and the penalty values are very difficult to adjust. For these reasons, using a single-objective approach, our first change to the classic SPP/SCP models is to consider minimization of the total number of duties (named $\mathrm{SPP_{Num}}/\mathrm{SCP_{Num}}$ models) as the objective function. Moreover because, even among transportation operators that permit drivers in a reserve situation, one must control the number and duration of overcover situations to avoid having too many duties covering the same piece of work at the same time. In this alternative approach, consider that the number of duties covering each piece of work is limited by a constant called *Maxc*. *Maxc* represents the maximum number of duties involved in the same piece of work. The planners must define this parameter, but contrary to the previously mentioned penalties, this parameter has a realistic meaning to the planner so he or she can easily decide on its value. The planners assign a value to this parameter based on their experience, and it is usually no higher than three.

The first proposed model for the DSP, designated by $\mathrm{SCP_{Num}}$_Maxc, has the following formulation:

$$(\mathrm{SCP_{Num}\_Maxc}) \quad \mathrm{MIN} \quad \sum_{j=1}^{n} x_j \tag{7}$$

$$\text{s.t.:} \quad \sum_{j=1}^{n} a_{ij} x_j \geq 1 \quad (i \in M) \tag{8}$$

$$\sum_{j=1}^{n} a_{ij} x_j \leq Maxc \quad (i \in M) \tag{9}$$

$$x_j \in \{0, 1\} \quad (j \in N) \tag{10}$$

Constraint set (8) guarantees that all pieces of work are covered by at least one duty and constraint set (9) guarantees that pieces of work are covered by at most *Maxc* duties for the feasible schedules.

This $\mathrm{SCP_{Num}}$_Maxc model is more complex and difficult to solve than the $\mathrm{SCP_{Num}}$ (SCP minimizing the total number of duties) because of the additional set of constraints. However it includes some aspects that meet the practical needs of the planner, one being that the model only includes parameters that have a real meaning to the planner and limits the number of overcover situations. Many planners complain that schedules that have four or more pieces of work overcovered are, in real situations, difficult or impossible to implement and to manually change without destroying a greater part of the solution.

Another important opinion obtained from our direct work with planners, is that, in some situations, it is easier to adjust a schedule with some pieces of work not covered than to change a schedule with all pieces of work covered as may be obtained from the SCP model. The idea is to work with *Relaxed Models* that allow some undercovered pieces of work in final schedules, but the number of pieces of work not covered should obviously be controlled.

The following model description can be seen as a relaxation of the $\mathrm{SCP_{Num}}$_Maxc model. We called this model $\mathrm{SCP_{Num}}$_Maxc_t, where the number of undercover pieces of work is controlled by a value, t, defined by the user that represents the minimum percentage of the number of pieces of work that must be covered in a final schedule. Again, this parameter makes sense to the planner and he or she can easily define it on the strength of his or her experience.

To present the $\mathrm{SCP_{Num}}$_Maxc_t model one must define a new set of decision variables:

$$y_i = \begin{cases} 1 & \text{if work piece } i \text{ is covered by any duty on the schedule;} \\ 0 & \text{otherwise;} \end{cases} \qquad (i \in M).$$

Considering the $\mathrm{SCP_{Num}}$_Maxc_t model the DSP has the following formulation:

$$(\mathrm{SCP_{Num}\_Maxc\_t}) \quad \mathrm{MIN} \quad \sum_{j=1}^{n} x_j \tag{11}$$

$$\text{s.t.:} \quad y_i \le \sum_{j=1}^{n} a_{ij} x_j \le Maxc \quad (i \in M) \tag{12}$$

$$\sum_{i=1}^{m} y_i \ge t \times |M| \tag{13}$$

$$x_j, y_i \in \{0, 1\} \quad (i \in M; j \in N) \tag{14}$$

Constraint set (12) guarantees that all pieces of work (if covered) are covered by at least one and at most by *Maxc* duties and constraint set (13) guarantees that the percentage of covered pieces of work is higher than t.

The next and last model reflects an important comment by the planners who consider that not all work duties are equal. The possibility of undercover or overcover situations can differ. depending on the type of the piece of work related to the hour and line of the passengers' trips Moreover in a large set of schedules it is not unusual for the overcover situation, consisting of more than four duties simultaneously covering the same piece of work, to occur in the middle of the day whereas, in relaxed models, undercover pieces of work occur at the beginning or end of the daily periods. To control this situation we propose an alternative approach that considers a division of M pieces of work set into two subsets, M^{SCP} and M^{SPP} ($M^{\mathrm{SCP}} \cup M^{\mathrm{SPP}} = M$ and $M^{\mathrm{SCP}} \cap M^{\mathrm{SPP}} = \{\}$). The pieces of work included in each subset will have different constraints related to permission or otherwise for overcover situations.

In this alternative model, called $\mathrm{SCP_{Num}}$_Maxc_t_$\mathrm{P_{scp}}$, the subsets idea is applied to the previous $\mathrm{SCP_{Num}}$_Maxc_t model. In this approach the DSP has the following formulation:

$$(\mathrm{SCP_{Num}\_Maxc\_t\_P_{scp}}) \quad \mathrm{MIN} \quad \sum_{j=1}^{n} x_j \tag{15}$$

$$\text{s.t.:} \quad y_i \leq \sum_{j=1}^{n} a_{ij} x_j \leq Maxc \quad (i \in M^{\text{SCP}}) \tag{16}$$

$$\sum_{j=1}^{n} a_{ij} x_j = y_i \quad (i \in M^{\text{SPP}}) \tag{17}$$

$$\sum_{i=1}^{m} y_i \geq t \times |M| \tag{18}$$

$$x_j, y_i \in \{0, 1\} \quad (i \in M; j \in N) \tag{19}$$

Constraint set (16) guarantees that all M^{SCP} pieces of work, if covered, are covered by at least one and at most by *Maxc* duties, constraint set (17) guarantees that all M^{SPP} pieces of work, if covered, are covered only by one duty and constraint set (18) guarantees that the percentage of covered pieces of work is higher than t. Consider the value P_{scp}, that represents the percentage of pieces of work in the subset, M^{SCP}, i.e. $(|M^{\text{SCP}}| = \lfloor |M| \times P_{scp} \rfloor)$. The set M^{SCP} can be defined by the planner or, given P_{scp}, it can be generated randomly.

6 Results

The next four phases of the methodology followed, Verify the Model; Select the Best Alternative; Present the Results of the Analysis and Implement and Evaluate are presented in this section. As mentioned earlier, this work was performed with the full collaboration of the end-users and planners from the companies concerned. The output schedules could be visualized by the planners who decided as to their quality and applicability.

To test the models proposed, we selected a set of DSP instances from the leading Portuguese transportation operators. For confidentiality reasons we are allowed to show only a subset of all the instances seen in practice. The instances selected endeavour to cover the entire set of different real situations found in the national passenger bus transportation business. We considered nine instances from three different transportation operators employing different planning strategies. The instances selected represent the Portuguese operators. In Table 1, we summarize the main characteristics of the operators chosen, which include:

- Urban and Suburban operators;
- Public and Private operators;
- "Family" operators that have a few drivers and big operators with a strong union force;
- Vehicle schedules with a different number and duration of vehicle blocks;
- Vehicle schedules with different relief points and number of pieces of work;
- Different types of duties with different rules to define them;
- Different planning strategies.

All feasible duties, built on the selected vehicle schedules, the piece of work division of the vehicle blocks and the set of parameters that reflect the duty rules, are

Table 1 The main characteristics of the operators selected

Properties		Operator		
		CARRIS	STCP	RL
Kind of Operator	Urban	*	*	
	Suburban			*
	Public	*	*	
	Private			*
	Small			*
	Big	*	*	
Planning Type	"Line by Line"	*	*	
	"By network"			*
Vehicles number average		10	9	33
Relief points average		2	3	11
Relief opportunities average		166	136	315
Different duty types average		3	5	4
Planning with extra work		No	No	Yes
Meal break limitation		Lunch and Dinner	Lunch	No
Obligation to start/end at the depot		No	Yes	No
Day periods not allowed to start/end duties		Yes	Yes	No

Table 2 The dimension of the test instances

	Carr1	Carr2	Carr3	STCP1	STCP2	STCP3	RL1	RL2	RL3
m	131	148	179	96	88	184	129	302	347
n	1879	13872	23747	2015	11145	42496	1839	5271	23305

generated on the basis of the work of Agra (1993). The method proposed by Agra is based on complete generation restricted to the specific rules of each company collaborating in this project. The method is applied only once to obtain all feasible pieces of work, so time is not a relevant issue. There are three different instances for each operator covering the usual dimension of the real DSP, which the planners should solve; see Table 2.

The three alternative models proposed, $\mathrm{SCP_{Num}}$_Maxc, $\mathrm{SCP_{Num}}$_Maxc_t and $\mathrm{SCP_{Num}}$_Maxc_t_$\mathrm{P_{scp}}$, were tested with Cplex 8.1 using parameters $Maxc = 3$, $t = 90\%$ and $P_{scp} = 0.5$ to obtain the optimal solution for the nine instances. Cplex 8.1 is commercial software for Integer Programming Resolution from ILOG.

The summary of the results obtained are displayed in Table 3. We solve each instance of the different companies considered and present the results of the different evaluation criteria obtained when solving each model proposed. It should be noted that the objective function may vary according to the different models and that the value of each evaluation criterion for each solution obtained is also given.

The models considered were (as indicated in the columns of the table):

- SCP/SPP (Classical Set Covering/Partitioning Model with objective function as a cost function);
- $\text{SCP}_{\text{Num}}/\text{SPP}_{\text{Num}}$ (Classical Set Covering/Partitioning Model with objective function minimizing the total number of duties).

And the new proposed and tested models:

- SCP_{Num}_3 (Model SCP_{Num}_Maxc with $Maxc = 3$ and objective function minimizing the total number of duties);
- SCP_{Num}_3_0.9 (Model SCP_{Num}_Maxc_t with $Maxc = 3$, $t = 90\%$ and objective function minimizing the total number of duties);
- SCP_{Num}_3_0.9_0.5 (SCP_{Num}_Maxc_t_P_{scp} with $Maxc = 3$, $t = 90\%$, $P_{scp} = 0.5$ and objective function minimizing the total number of duties).

The value that each solution presents for the different evaluation criteria is indicated in the respective cell of the row of each evaluation criterion and the column of the model that obtains that solution. The evaluation criteria considered (presented in Sect. 4) were:

(1) Total "real" cost;
(2) Total number of undercover pieces of work;
(3) Total duration of undercover pieces of work;
(4) Total number of overcover pieces of work;
(5) Total duration of overcover pieces of work;
(6) Maximum number of duties involved in the same overcover piece of work;
(7) Unfit;
(8) Total number of duties in the optimal schedule;
(9) Number of trippers.

The shaded values on the table correspond to cases where the Cplex 8.1 was interrupted after 24 hours running without obtaining any final solution or models which came up with no feasible solution.

When we compare the results of the new model SCP_{Num}_3 with the SCP_{Num} model, we find that:

- The number of duties are equal (evaluation criterion (8));
- The number of pieces of work involved in overcover situations also undergoes small variations (evaluation criterion (4));
- But the maximum number of duties covering the same row at the same time (evaluation criteria (6)) improved for the suburban instances (RL1, RL2 and RL3).

Apparently, the model SCP_{Num}_3 leads to results equal to the SCP_{Num} model with respect to the values obtained for all evaluation criteria except number (6). However, when presenting these results to planners they comment that it is very important that the maximum number of duties cover a piece of work under a specific value, even if some other evaluation criteria increase, especially for suburban instances. Therefore, the planners find it extremely useful to be able to obtain the solutions from both models as well as evaluate and compare them in order to make a better decision. For example, in the RL2 case where evaluation criterion "maximum number of duties involved in the same overcover piece of work" (6) is larger for the SCP_{Num} model

Table 3 Value of the optimal schedules for the different evaluation criteria

Instances		SCP	SCP_{Num}	SPP	SPP_{Num}	SCP_{Num_3}	$SCP_{Num\_3\_0.9}$	$SCP_{Num\_3\_0.9\_0.5}$
Carr1	(1)	214975	320966			329544	190392	181460
	(2)	0	0			0	13	14
	(3)	0	0			0	1504	1687
	(4)	23	21			23	3	6
	(5)	3212	2691			2700	3	552
	(6)	4	4			3	2	2
	(7)	37	29			29	16	20
	(8)	38	38			38	26	26
	(9)	3	3			3	0	0
Carr2	(1)	91442	182148			218220	104036	104144
	(2)	0	0			0	15	14
	(3)	0	0			0	712	630
	(4)	32	23			19	0	1
	(5)	1551	1119			1008	0	53
	(6)	2	3			3	1	2
	(7)	32	24			21	15	14
	(8)	19	19			19	14	14
	(9)	0	0			0	0	0
Carr3	(1)	117628	226372	180822	190398	226372	101644	101803
	(2)	0	0	0	0	0	18	17
	(3)	0	0	0	0	0	1187	1070
	(4)	41	14	0	0	14	7	5
	(5)	1852	694	0	0	694	316	191
	(6)	3	2	1	1	2	2	2
	(7)	46	14	0	0	14	25	22
	(8)	24	22	22	22	22	17	17
	(9)	0	0	0	0	0	0	0
STCP1	(1)	197434	276164	197434	268065	276164	284391	284078
	(2)	0	0	0	0	0	10	8
	(3)	0	0	0	0	0	1145	925
	(4)	0	6	0	0	6	3	0
	(5)	0	615	0	0	615	249	0
	(6)	1	2	1	1	2	2	1
	(7)	0	6	0	0	6	13	8
	(8)	25	21	25	21	21	17	17
	(9)	0	0	0	0	0	0	0
STCP2	(1)	94914	132791	94914	124958	132791	158199	158245
	(2)	0	0	0	0	0	7	6
	(3)	0	0	0	0	0	353	305
	(4)	0	7	0	0	7	6	5
	(5)	0	343	0	0	343	300	252
	(6)	1	2	1	1	2	2	2

Table 3 (*Continued*)

Instances		SCP	SCP_{Num}	SPP	SPP_{Num}	SCP_{Num_3}	$SCP_{Num\_3\_0.9}$	$SCP_{Num\_3\_0.9\_0.5}$
	(7)	0	7	0	0	7	13	11
	(8)	10	9	10	9	9	8	8
	(9)	0	0	0	0	0	0	0
STCP3	(1)		249755		264753	249755	279349	284119
	(2)		0		0	0	18	19
	(3)		0		0	0	959	997
	(4)		5		0	5	2	2
	(5)		210		0	210	100	100
	(6)		2		1	2	2	2
	(7)		5		0	5	20	21
	(8)		18		18	18	15	15
	(9)		0		0	0	0	0
RL1	(1)	297615	324307	377264	452913	312479	224927	226326
	(2)	0	0	0	0	0	13	13
	(3)	0	0	0	0	0	1633	1558
	(4)	16	25	0	0	28	12	12
	(5)	1534	2345	0	0	2845	917	896
	(6)	3	3	1	1	3	2	2
	(7)	20	29	0	0	33	25	25
	(8)	30	29	36	35	29	20	21
	(9)	1	2	5	11	1	0	0
RL2	(1)		893060		1099083	928443	629147	648497
	(2)		0		0	0	31	31
	(3)		0		0	0	6977	6332
	(4)		65		75	81	44	28
	(5)		6610		8123	11273	3389	2650
	(6)		7		4	4[a]	3	3
	(7)		104		117	138	84	67
	(8)		93		93	93	62	66
	(9)		1		9	1	0	0
RL3	(1)		517158					
	(2)		0					
	(3)		0					
	(4)		36					
	(5)		2460					
	(6)		3					
	(7)		38					
	(8)		48					
	(9)		0					

[a] For $Maxc = 3$ there is no feasible solution, so this solution was obtained for $Maxc = 4$

then for the model SCP_{Num}_4, the solution obtained by this last one model is clearly better from the planners point of view, since both solutions have the same number of duties.

As for the SCP_{Num}_3_0.9 and SCP_{Num}_3_0.9_0.5 models, it makes no sense to compare them with the previous cases since the latter allow undercover pieces of work. So the other evaluation criteria are implicitly improved. When we compare them, we find that both models have similar, balanced solutions with respect to the different evaluation criteria. We presented these solutions to the planners, who commented that these schedules are easier to adjust and modify in a shorter time than other schedules obtained by the previous models. They prefer to have partial solutions which are easier to adjust, rather than complete solution with too many duties and overcover situations.

In summarising, for urban DSP instances, such as the ones from Carr and STCP, the SCP_{Num}_3 model works quite well, the number of overcover is under control and the solutions obtained can be implemented in real situations without a significant number of modifications. However, for the suburban instances, such as the RL instances, the models SCP_{Num}_3_0.9 and SCP_{Num}_3_0.9_0.5 are the ones the planners prefer. The classic models for these instances are very difficult to solve, whereas the new models are solved quite easily. Moreover, the fact that one is able to manually modify the schedules obtained by these new models in a short period of time is, according to the planners the most important characteristic. The three new models are quite simple since they are based on modifications of the well know classic SCP/SPP models. Also, the optimal solution for these new proposed models can be obtained using commercial software in most cases. It should be remembered that these models are closer to the real situation phased by the planner, as their comments and suggestions were born in mind before the models were created. Therefore, the new models we propose enjoy the advantage of better applicability, simplicity and quality of solutions similar to the classic SPP/SCP based models.

7 Conclusions

In this work we have considered the Driver Scheduling Problem in a realistic context. Our objective was to propose new mathematical models for this problem, models that can more closely represent the complexity and issues present in the planning process of the transportation companies. Development of these models was based on broad collaboration with different bus transportation companies in Portugal.

We analysed and evaluated the different alternative models for the DSP. We proposed several alternative models and the corresponding optimal schedules were analysed and criticised by the end-users and planners of the transportation companies involved. The models and respective schedules were evaluated by the planners with respect to their quality, which was measured in terms of simplicity, solution quality and applicability of the models.

The principal conclusion of this research work is that the models proposed are more flexible and adaptable to the different realities and evaluation criteria for each of the different transportation operators. The planners consider that the schedules obtained from the new models are more balanced with respect to the global set of evaluation criteria. They add that even when the schedules are not final and directly applied to the real situation, they are easier to modify and adjust.

This study reveals that the new alternative models are, from a practical point of view better for the DSP solving process. We intend to continue this modelling approach to the DSP, pursue the work already done and consider a multiobjective approach. At the same time, we intend to develop and implement new metaheuristic techniques to obtain a solution for the models proposed insofar that many of the large instances can not be solved in a reasonable timeframe by general ILP software.

Acknowledgements Thanks are due to Fundação para a Ciência e Tecnologia (SFRH/ BD / 1054 / 2000) for providing funding to support the work described in this paper for the first author. Thanks are due to Ministerio de Educación y Ciencia, Spain (SEJ2006-12291/ECON) for providing funding to support the work of the second author.

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